

Teacher Information, Course Description and  
Lecture-wise Plan  
of  
Operating Systems

under

Pilot Project:  
Uniform Teaching, Assessment and Quality  
Assurance Across All The Campuses

by

Faculty of Information Science and Technology  
COMSATS University Islamabad

**A: Teacher Information**

|                                |                     |
|--------------------------------|---------------------|
| Course Name: Operating Systems | Credit Hours:       |
| Course Code:(CSC323)           | Pre-requisite:None  |
| Teacher Name:                  | E-mail:             |
| Program: BS (Computer Science) | Semester: Fall 2026 |
| Department: Computer Science   | Campus:             |

**B: Course Description****Topics Covered:**

This course introduces the services and functions performed by operating system for smooth and accurate system operations. Topics include: Operating Systems Overview; Device Organization & System Operations; Operating Systems Principles; Process Management; Process Synchronization; Deadlocks; Multiprocessor Issues; Memory Management; Storage Management; Security & Protection; case studies of desktop and mobile systems.

**Textbook:**

1. Operating System Concepts, Silberschatz & Galvin, Addison-Wesley, 2021.

**Reference Book:**

2. Operating Systems: Internals and Design Principles, Stallings, W., Pearson, 2017.
3. Modern Operating Systems, Tanenbaum, A. S., Prentice Hall, 2014.

### C: Lecture-wise Activities

| Week # | Lecture #   | Topic (Chapter/Topics Covered)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Evaluation   | Remarks              |
|--------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| 1      | 1           | <b>Overview of Operating System:</b> <ul style="list-style-type: none"> <li>Operating system <ul style="list-style-type: none"> <li>User View</li> <li>System View</li> </ul> </li> <li>Computer System Organization <ul style="list-style-type: none"> <li>Interrupts and its implementation</li> <li>I/O Structure</li> </ul> </li> <li>Computer System Architecture <ul style="list-style-type: none"> <li>Single processor System</li> <li>Multiprocessor system</li> <li>Clustered System</li> </ul> </li> <li>Operating system operations <ul style="list-style-type: none"> <li>Multiprogramming &amp; Multitasking</li> <li>Dual mode and Multimode</li> </ul> </li> </ul> |              | Silberschatz:<br>Ch1 |
|        | 2           | <b>Device Organization &amp; System Operations:</b> <ul style="list-style-type: none"> <li>Computing Environments <ul style="list-style-type: none"> <li>Traditional</li> <li>Mobile</li> <li>Client-Server</li> <li>Peer to peer</li> <li>Cloud</li> </ul> </li> <li>Interrupts <ul style="list-style-type: none"> <li>Interrupt-driven operation;</li> <li>Types of Interrupt</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                             |              | Silberschatz:<br>Ch1 |
| 2      | 3<br>(Date) | <b>Operating System Structure&amp; Services:</b> <ul style="list-style-type: none"> <li>OS services</li> <li>User and OS Interfaces <ul style="list-style-type: none"> <li>Command Interface</li> <li>Graphical User Interface</li> <li>Mobile</li> <li>Touch Screen</li> <li>Choice of Interface</li> </ul> </li> <li>System calls</li> </ul>                                                                                                                                                                                                                                                                                                                                     |              | Silberschatz:<br>Ch2 |
|        | 4<br>(Date) | <b>OS Structure&amp;Design Issues:</b> <ul style="list-style-type: none"> <li>OS Design and Implementation</li> <li>OS Structure <ul style="list-style-type: none"> <li>Monolithic</li> <li>Layered</li> <li>Microkernel</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    | Assignment#1 | Silberschatz:<br>Ch2 |

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|   |              | <ul style="list-style-type: none"> <li>○ Modular</li> <li>○ Hybrid Syste</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                 |                          |
| 3 | 5<br>(Date)  | <ul style="list-style-type: none"> <li>• OS security and Protection mechanism</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Quiz # 1</b>                 | Silberschatz:<br>Ch16&17 |
|   | 6<br>(Date)  | <b>Process Management:</b> <ul style="list-style-type: none"> <li>• Process concept <ul style="list-style-type: none"> <li>○ Process State</li> <li>○ PCB</li> </ul> </li> <li>• Process Scheduling <ul style="list-style-type: none"> <li>○ Scheduling Queues</li> <li>○ Process Scheduling</li> <li>○ Context Switch</li> </ul> </li> <li>• Process Operations <ul style="list-style-type: none"> <li>○ Process Creation (Fork, Exec &amp; Wait System-Calls)</li> <li>○ Process Termination (Exit &amp; abort system calls, Cascading, Termination, and Zombie &amp; Orphan Processes)</li> </ul> </li> </ul> |                                 | Silberschatz:<br>Ch3     |
| 4 | 7<br>(Date)  | <b>Inter-Process Communication:</b> <ul style="list-style-type: none"> <li>• Inter-process communication (IPC) <ul style="list-style-type: none"> <li>○ IPC in shared memory</li> <li>○ IPC in message Passing</li> </ul> </li> <li>• Sockets(Remote Procedure Call)</li> </ul>                                                                                                                                                                                                                                                                                                                                  | Exercises as given in the books | Silberschatz:<br>Ch3     |
|   | 8            | <b>Threads and Concurrency:</b> <ul style="list-style-type: none"> <li>• Threads(Overview)</li> <li>• Processes vs. Threads,</li> <li>• Multiple Simultaneous Computations</li> <li>• Alhamad's Law</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                   |                                 | Silberschatz:<br>Ch4     |
| 5 | 9<br>(Date)  | <b>Threads and Concurrency:</b> <ul style="list-style-type: none"> <li>• Multicore Programming &amp; its Challenges</li> <li>• Multi Treaded Models</li> <li>• Multi Treaded Libraries</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Assignment#2</b>             | Silberschatz:<br>Ch4     |
|   | 10<br>(Date) | <b>CPU Scheduling:</b> <ul style="list-style-type: none"> <li>• CPU burst/I/O burst</li> <li>• CPU scheduler</li> <li>• Preemptive vs non-Preemptive</li> <li>• Dispatcher</li> <li>• Scheduling Criteria</li> <li>• FCFS</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             |                                 | Silberschatz:<br>Ch5     |
| 6 | 11           | <b>Scheduling Algorithms:</b> <ul style="list-style-type: none"> <li>• SJF</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 | Silberschatz:<br>Ch5     |

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|   | (Date)       | <ul style="list-style-type: none"> <li>• <i>SRTF</i></li> <li>• <i>Round Robin</i></li> <li>• <i>Time quantum and context switch</i></li> <li>• <i>Priority scheduling</i></li> <li>• <i>Priority scheduling vs Round Robin</i></li> </ul>                                                                                                |                          |                              |
|   | 12<br>(Date) | <b>Scheduling Algorithms:</b> <ul style="list-style-type: none"> <li>• <i>Multilevel Queue</i></li> <li>• <i>Multilevel Feedback Queue</i></li> <li>• <i>Thread Scheduling, Pthread Scheduling</i></li> <li>• <i>Discussion on scheduling algorithms used in windows and linux</i></li> </ul>                                             | <b>Quiz # 2</b>          | <i>Silberschatz:<br/>Ch5</i> |
| 7 | 13<br>(Date) | <b>Synchronization:</b> <ul style="list-style-type: none"> <li>• <i>Race Condition</i></li> <li>• <i>Critical section problem</i></li> <li>• <i>Interrupt based solution vs software solution</i></li> <li>• <i>Petersons solution</i></li> <li>• <i>Petersons solution revisited with memory barrier</i></li> </ul>                      |                          | <i>Silberschatz:<br/>Ch6</i> |
|   | 14<br>(Date) | <b>Process Synchronization:</b> <ul style="list-style-type: none"> <li>• <i>Hardware instructions</i></li> <li>• <i>Test &amp; set, compare &amp; swap</i></li> <li>• <i>Atomic Variables</i></li> <li>• <i>Mutex Locks</i></li> <li>• <i>Semaphores</i></li> <li>• <i>Semaphore implementation</i></li> <li>• <i>Monitors</i></li> </ul> |                          | <i>Silberschatz:<br/>Ch6</i> |
| 8 | 15<br>(Date) | <b>Classical Synchronization Problems:</b> <ul style="list-style-type: none"> <li>• <i>Bounded Buffer Problem</i></li> <li>• <i>Readers Writers Problem</i></li> </ul>                                                                                                                                                                    |                          | <i>Silberschatz:<br/>Ch7</i> |
|   | 16           | <b>Classical Synchronization Problems:</b> <ul style="list-style-type: none"> <li>• <i>Dining Philosophers problem</i></li> <li>• <i>Monitors solution to Dining Philosophers</i></li> <li>• <i>Synchronization in windows and linux.</i></li> </ul>                                                                                      |                          | <i>Silberschatz:<br/>Ch7</i> |
| 9 | 17<br>(Date) | <i>Midterm Examination</i>                                                                                                                                                                                                                                                                                                                | <b>Midterm<br/>(25%)</b> |                              |

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|    | 18<br>(Date) | <i>Midterm Examination</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                        |                                     |
| 10 | 19<br>(Date) | <b>Deadlocks:</b> <ul style="list-style-type: none"> <li>• <i>Deadlock: Definition and real-world/system examples</i></li> <li>• <i>Necessary Conditions for Deadlock: Mutual Exclusion, Hold-and-Wait, No Preemption, Circular Wait</i></li> <li>• <i>Resource-Allocation Graph (RAG): Notation, request edges, assignment edges</i></li> <li>• <i>Interpreting Cycles in a RAG (cycle vs. guaranteed deadlock, single vs. multiple instances)</i></li> </ul>                                                                | <b>Assignment#3</b>                    | <i>Silberschatz:<br/>Ch8</i>        |
|    | 20<br>(Date) | <b>Deadlock Avoidance:</b> <ul style="list-style-type: none"> <li>• <i>Deadlock Avoidance: Concept of Safe State and Safe Sequence</i></li> <li>• <i>Resource-Allocation Graph Algorithm Banker's Algorithm (multiple instances per resource type)</i></li> </ul>                                                                                                                                                                                                                                                             | <i>Exercises as given in the books</i> | <i>Silberschatz:<br/>Ch8</i>        |
| 11 | 21<br>(Date) | <b>Deadlock Detection, Recovery &amp; Case Study:</b> <ul style="list-style-type: none"> <li>• <i>Deadlock Detection: Single instance per resource type : Wait-for Graph and cycle detection</i></li> <li>• <i>Deadlock Detection: Multiple instances per resource type: Detection Algorithm (Available, Allocation, Request)</i></li> <li>• <i>Detection-Algorithm Usage</i></li> <li>• <i>Recovery from Deadlock</i></li> <li>• <i>Case Study: Process synchronization &amp; deadlock handling in Windows OS</i></li> </ul> |                                        | <i>Silberschatz:<br/>Ch8&amp;21</i> |
|    | 22<br>(Date) | <b>Memory Management:</b> <ul style="list-style-type: none"> <li>• <i>Background: Physical Memory and Memory Management Hardware (</i></li> <li>• <i>Base and Limit Registers, Memory Protection</i></li> <li>• <i>Address Binding: Compile-Time, Load-Time, and Execution-Time binding</i></li> <li>• <i>Logical vs. Physical Address Space; the MMU and the Relocation</i></li> </ul>                                                                                                                                       | <b>Quiz # 3</b>                        | <i>Silberschatz:<br/>Ch9</i>        |

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|    |              | <i>Register</i> <ul style="list-style-type: none"> <li>• <i>Dynamic Loading: loading routines only when called</i></li> <li>• <i>Dynamic Linking and Shared Libraries</i></li> <li>• <i>Windows Example: DLL (Dynamic Link Library) loading and shared-library resolution at run time</i></li> </ul>                                                                                                                                                                                                                                                                                                          |  |                       |
| 12 | 23<br>(Date) | <b>Contiguous Memory Allocation &amp; Segmentation:</b> <ul style="list-style-type: none"> <li>• <i>Contiguous Memory Allocation: Fixed Partitioning (MFT)</i></li> <li>• <i>Contiguous Memory Allocation: Dynamic Partitioning (MVT)</i></li> <li>• <i>Allocation Strategies: First-Fit, Best-Fit, and Worst-Fit</i></li> <li>• <i>Fragmentation: Internal vs. External, and Compaction</i></li> <li>• <i>Non-Contiguous Allocation: Segmentation: basic method and user's view of memory</i></li> <li>• <i>Segmentation Hardware: Segment Table, Base and Limit per segment, protection bits</i></li> </ul> |  | Silberschatz:<br>Ch9  |
|    | 24<br>(Date) | <b>Paging: Basic Method &amp; Hardware Support:</b> <ul style="list-style-type: none"> <li>• <i>Paging: Basic Method &amp; Hardware Support</i></li> <li>• <i>Paging: Basic Method: Page Table, Frame Table, address translation</i></li> <li>• <i>Hardware Support: Translation Look-aside Buffer (TLB) and Effective Access Time</i></li> <li>• <i>Protection in Paging: Valid-Invalid Bit, Read/Write/Execute permission bits</i></li> <li>• <i>Advanced: Hashed Page Tables and Inverted Page Tables</i></li> </ul>                                                                                       |  | Silberschatz:<br>Ch9  |
| 13 | 25<br>(Date) | <b>Virtual Memory &amp; Demand Paging:</b> <ul style="list-style-type: none"> <li>• <i>Virtual Memory: Background and Benefits over pure physical-memory schemes</i></li> <li>• <i>Demand Paging: Basic Concept and Pure Demand Paging</i></li> <li>• <i>Page-Fault Handling: step-by-step sequence of events</i></li> </ul>                                                                                                                                                                                                                                                                                  |  | Silberschatz:<br>Ch10 |

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|    |              | <ul style="list-style-type: none"> <li>• <i>Performance of Demand Paging: computing Effective Access Time (EAT)</i></li> <li>• <i>Copy-on-Write: sharing pages between parent and child processes</i></li> </ul>                                                                                     |                                               |                                  |
|    | 26<br>(Date) | <b>Page Replacement Algorithms:</b> <ul style="list-style-type: none"> <li>• <i>Overview</i></li> <li>• <i>FIFO</i></li> <li>• <i>Optimal</i></li> <li>• <i>LRU</i></li> <li>• <i>SecondChance Algorithm</i></li> <li>• <i>MFU &amp; LFU</i></li> <li>• <i>Thrashing.</i></li> </ul>                 | <i>Exercises / page-reference strings</i>     | <i>Silberschatz: Ch10</i>        |
| 14 | 27<br>(Date) | <b>Case Studies:</b> <ul style="list-style-type: none"> <li>• <i>Memory Management in Windows and on ARM architecture</i></li> </ul>                                                                                                                                                                 | <i>Assignment#4</i>                           | <i>Silberschatz: Ch9&amp;21</i>  |
|    | 28<br>(Date) | <b>Mass Storage Management:</b> <ul style="list-style-type: none"> <li>• <i>Hard disks</i></li> <li>• <i>Volatile and non-volatile memory</i></li> <li>• <i>Disk scheduling</i></li> <li>• <i>NVM scheduling</i></li> </ul>                                                                          | <i>Quiz # 4</i>                               | <i>Silberschatz: Ch11</i>        |
| 15 | 29<br>(Date) | <b>Storage Device Management:</b> <ul style="list-style-type: none"> <li>• <i>Formatting</i></li> <li>• <i>Boot Block</i></li> <li>• <i>Bad Blocks; and Swap-Space Management.</i></li> </ul>                                                                                                        |                                               | <i>Silberschatz: Ch11</i>        |
|    | 30<br>(Date) | <b>File System:</b> <ul style="list-style-type: none"> <li>• <i>File concept</i></li> <li>• <i>File attributes and operations</i></li> <li>• <i>Access methods</i></li> <li>• <i>Directory structure</i></li> <li>• <i>File-System Structure</i></li> <li>• <i>File System Operations</i></li> </ul> |                                               | <i>Silberschatz: Ch13</i>        |
| 16 | 31<br>(Date) | <b>File System:</b> <ul style="list-style-type: none"> <li>• <i>Directory Implementation</i></li> <li>• <i>File Protection; and Memory-Mapped Files.</i></li> </ul>                                                                                                                                  |                                               | <i>Silberschatz: Ch13</i>        |
|    | 32<br>(Date) | <b>Case studies:</b> <ul style="list-style-type: none"> <li>• <i>Windows and Apple file system</i></li> </ul>                                                                                                                                                                                        | <i>Final Review / Examination Preparation</i> | <i>Silberschatz: Ch14&amp;21</i> |
| 17 | (Date)       |                                                                                                                                                                                                                                                                                                      | <b>Final (50%)</b>                            |                                  |